

About Simulation Lifecycle & Resimulation Standard - (SLRS)

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Architecture: SIMULOS® — Simulation Governance Architecture Category:
Governance & Enforcement Subcategory: Simulation Governance Parent Standard:
Simulation Lifecycle & Resimulation Standard (SLRS) Governed Space: Simulation
Lifecycle, Drift & Resimulation

Canonical Definition

Simulation Lifecycle & Resimulation Standard (SLRS) governs how the continuing relevance of a simulation is monitored after its creation and use, how material divergence between the governed simulation basis and changing reality is recognized, and when change requires simulation update, resimulation, replacement, or retirement.

SLRS addresses a fundamental lifecycle problem:

A simulation that was appropriate when created may become progressively less representative as the object, environment, assumptions, data, operational conditions, or reality itself changes.

The Standard prevents historical simulation results from retaining unchanged authority after the conditions that gave them meaning have materially changed.

Why SLRS Matters

Simulation governance does not end when a simulation campaign finishes.

Systems evolve. Environments change. New operational data appears. Previously unknown failure modes emerge. Models improve. Assumptions become obsolete. Inputs shift. New scenarios become relevant.

SLRS establishes a governed mechanism for asking:

Is this simulation still relevant to the reality for which it is being relied upon?

Where the answer changes materially, SLRS determines whether the appropriate response is:

continued use → review → update → partial resimulation → broader resimulation
→ replacement → retirement.

Governance Flow

Governed Simulation Evidence & Reliance Basis → Lifecycle Monitoring →
Simulation Drift Recognition → Resimulation Trigger → Resimulation Scope
Determination → Resimulation / Replacement / Retirement → Updated Governed

Simulation Lifecycle

This creates the recursive lifecycle:

Simulation → Operational Use → Reality Change → Simulation Drift →
Resimulation Trigger → Resimulation / Replacement / Retirement What SLRS Helps
Organizations Govern

SLRS helps organizations prevent simulation results from becoming
methodologically stale while remaining operationally influential.

It provides a structured basis for monitoring changes in: the Simulation
Object; object configuration and version; represented reality; environmental
conditions; assumptions; models; inputs and data; scenarios; observed
operational behavior; simulation-to-reality correlation; evidence relevance;
known limitations.

Not every change requires complete resimulation.

SLRS exists partly to determine which changes are material enough to require
action and how much of the simulation lifecycle must be reopened. Use Case 1 —
Space Mission: Mission Reality Changes After Simulation

A space mission has undergone extensive simulation covering trajectory,
spacecraft behavior, navigation, landing conditions, environmental effects,
system failures, crew procedures, and mission contingencies.

Months later, new information becomes available.

The spacecraft configuration has changed. Updated environmental observations
alter expected surface conditions. A component behaves differently during
physical testing. New mission constraints modify the landing profile.

The original simulations may still be technically correct representations of
the conditions under which they were executed.

But those conditions are no longer identical to the mission now being
prepared.

The critical governance question becomes:

Which previous simulation results remain relevant, which have materially
drifted from the current mission basis, and what must be simulated again?

SLRS provides the lifecycle mechanism for identifying those changes,
determining their materiality, activating appropriate resimulation triggers,
and defining whether resimulation must cover one affected scenario, one
subsystem, a scenario family, or a much larger part of the mission simulation
campaign.

This prevents two opposite failures:

continuing to rely on obsolete simulations

and

unnecessarily repeating an entire simulation program whenever anything
changes.

Resimulation should follow material change, not merely the passage of time.

Use Case 2 — Autonomous Vehicle or Robot: Operational Reality Evolves

An autonomous vehicle or robot is developed through millions of simulated scenarios and subsequently enters real-world operation.

Operational experience begins producing conditions that were absent or materially different in the original simulation environment.

New road layouts appear. Sensor hardware changes. Software behavior evolves. Human interaction patterns differ from those represented during development. Previously rare edge cases become observable. Weather, lighting, surfaces, obstacles, or environmental distributions change.

The original simulation campaign does not suddenly become useless.

But neither should its relevance be assumed to remain unchanged.

SLRS establishes continuous lifecycle governance between:

the simulation basis that existed when evidence was produced

and

the operational reality that exists now.

Material differences become Simulation Drift Candidates.

Where drift becomes significant, SLRS can trigger targeted resimulation of the affected conditions rather than automatically rebuilding the entire simulation campaign.

If the underlying model or representation has become fundamentally unsuitable, the appropriate outcome may instead be replacement or retirement.

This creates a feedback loop:

Real-World Operation → New Reality Evidence → Drift Recognition → Resimulation
→ Updated Simulation Basis

The simulation therefore becomes a governed evolving representation rather than a static artifact frozen at development time.

A simulation should not remain authoritative merely because it was once representative.

Enterprise Value

SLRS enables organizations to maintain simulation relevance across long operational lifecycles without treating every change as either irrelevant or grounds for complete resimulation.

It supports:

change-sensitive simulation governance → proportionate resimulation →
preserved relevance → controlled retirement.

For organizations operating large simulation programs, this can reduce unnecessary computational and engineering repetition while making material

simulation drift visible before outdated simulation evidence continues influencing important decisions.

Foundational Principle

A simulation remains governably relevant only while the conditions that support its meaning remain materially applicable; when those conditions change, continued reliance requires governed reassessment.

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Canonical Source

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